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Effects of precipitate shape and orientation on dispersion strengthening in magnesium alloys

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Abstract

This paper reports results on the development of the Orowan equation appropriate for magnesium alloys strengthened by rationally-oriented, shear-resistant precipitate plates/rods. Comparisons of identical volume fractions and number densities of precipitates per unit volume indicate that plate-shaped precipitates that form on prismatic planes of the magnesium matrix phase are most effective for dispersion strengthening.

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1. Introduction

Rationally-oriented, plate- and rod-shaped precipitates of metastable transition or equilibrium phases are often key strengthening constituents in many magnesium and aluminium alloys [1]. In precipitation-hardened magnesium alloys, for example, strengthening is attributable to plate-shaped precipitate phases formed on prismatic or basal planes of α -Mg matrix phase [1–10]. Although the shape, orientation and distribution of precipitate particles have long been recognised as potentially important factors in determining the strength of alloys, their quantitative effects on yield strength of precipitation-hardened magnesium alloys have received little attention. An

understanding of the quantitative relationship between yield strength and precipitate microstructures is currently limited by a lack of appropriate versions of the Orowan equation defining precipitation strengthening for such microstructures.

The present work seeks to develop appropriate versions of the Orowan equation for magnesium alloys containing rationally-oriented, shear-resistant precipitate plates/rods and to compare quantitatively the effects of the shape and orientation of precipitates on the critical resolved shear stress (CRSS) increment in such magnesium alloys. The scope of the present paper is restricted to plastic deformations that occur predominantly via the motion of dislocations gliding on the $(0001)_\alpha$ slip plane of the α -Mg matrix phase [11,12]. Interactions between shear-resistant precipitate plates/rods and dislocations gliding on $\{10\bar{1}2\}_\alpha$ twin planes of α -Mg [13,14] is the subject of a separate paper that will be reported elsewhere.

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2. Orowan strengthening

For point obstacles of identical strength, the Orowan increment in CRSS produced by the need for dislocations to by-pass these obstacles is given as [15–18]:

$$\Delta\tau = \frac{Gb}{2\pi\lambda\sqrt{1-\nu}} \ln \frac{d_p}{r_0}, \quad (1)$$

where $\Delta\tau$ is the increment in CRSS due to dispersion strengthening, G the shear modulus of the magnesium matrix phase, b the magnitude of the Burgers vector of the slip dislocations, ν the Poisson's ratio, λ is the effective planar inter-obstacle spacing, d_p the mean planar diameter of the point obstacles, and r_0 the core radius of dislocations. For the purpose of comparison, it is convenient to restrict the distribution of precipitate particles to a triangular array on the $(0001)_\alpha$ slip plane of the α -Mg matrix phase and to assume $r_0 = b$.

2.1. Spherical precipitates

For a triangular array of spherical particles of uniform diameter d_t , the effective planar inter-precipitate spacing is calculated as

$$\begin{aligned} \lambda &= L_p - d_p = \frac{1.075}{\sqrt{N_a}} - \frac{\pi d_t}{4} \\ &= \left(\frac{0.779}{\sqrt{f}} - 0.785 \right) d_t, \end{aligned} \quad (2)$$

where L_p is the mean planar centre-to-centre inter-precipitate spacing, N_a the number of particles intersected per unit area of slip plane, and f the volume fraction of precipitates. In Eq. (2) $N_a = f/S_p$, where S_p is the mean planar cross-sectional area of each single particle. Eq. (1) can thus be rewritten in the form:

$$\Delta\tau = \frac{Gb}{2\pi\sqrt{1-\nu} \left(\frac{0.779}{\sqrt{f}} - 0.785 \right) d_t} \ln \frac{0.785 d_t}{b}. \quad (3)$$

2.2. Prismatic precipitate plates

For precipitate plates of uniform diameter d_t and thickness t_t ($d_t \gg t_t$) that form on $\{10\bar{1}0\}_\alpha$ or

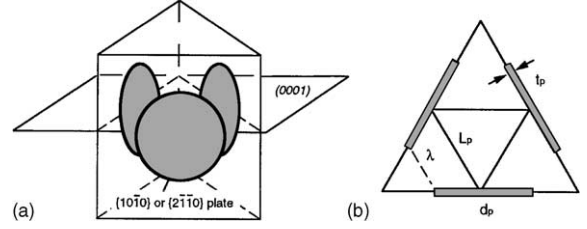


Fig. 1. (a) Arrangement of $\{10\bar{1}0\}_\alpha$ or $\{2\bar{1}\bar{1}0\}_\alpha$ precipitate plates in a triangular prismatic volume of α -Mg matrix, and (b) projection of intersected prismatic plates in $(0001)_\alpha$ slip plane of α -Mg.

$\{2\bar{1}\bar{1}0\}_\alpha$ prismatic planes of α -Mg, the angle between the habit plane of the precipitate plates and $(0001)_\alpha$ slip plane is 90° . The cross-section of the prismatic plates intersected in the slip plane is approximately rectangular in shape, defined by the mean planar thickness $t_p (= t_t)$ and the mean planar diameter $d_p (= \pi d_t/4)$, and has a mean planar area of $\pi d_t t_t/4$. The number density N_a of precipitates is given as $N_v d_t$ [19], where N_v represents the number density of precipitate plates per unit volume of the matrix phase. If the prismatic plates are assumed to have an ideal arrangement at the centre of each surface of a triangular volume in the matrix phase, Fig. 1(a), then the precipitate plates intersected by the $(0001)_\alpha$ slip plane will have a triangular array on that slip plane, Fig. 1(b). The effective planar inter-particle spacing is given as

$$\begin{aligned} \lambda &= L_p - \frac{d_p}{2} - \frac{\sqrt{3}t_p}{2} \\ &= 0.825\sqrt{\frac{d_t t_t}{f}} - 0.393d_t - 0.866t_t. \end{aligned} \quad (4)$$

Substituting Eq. (4) into Eq. (1), and assuming that the logarithmic term in Eq. (1) is $\ln(\sqrt{d_p t_p}/b)$, the appropriate version of the Orowan equation for magnesium alloys containing prismatic precipitate plates is given as

$$\begin{aligned} \Delta\tau &= \frac{Gb}{2\pi\sqrt{1-\nu} \left(0.825\sqrt{\frac{d_t t_t}{f}} - 0.393d_t - 0.886t_t \right)} \\ &\quad \times \ln \frac{0.886\sqrt{d_t t_t}}{b}. \end{aligned} \quad (5)$$

2.3. Basal precipitate plates

For plates of uniform diameter d_t and thickness t_t ($d_t \gg t_t$), and of $(0001)_\alpha$ habit plane, the probability that an $(0001)_\alpha$ plate is intersected by an $(0001)_\alpha$ plane of α -Mg is given as t_t [19]. The number density, N_a , of precipitate plates is thus $N_v t_t$. The mean planar diameter of the plates is equal to d_t , and the mean planar cross-sectional area is $\pi d_t^2/4$. Assuming a triangular array distribution of the $(0001)_\alpha$ plates in the slip plane of α -Mg, Fig. 2(a) and (c), the effective planar inter-particle spacing is given as

$$\lambda = L_p - d_p = \left(\frac{0.953}{\sqrt{f}} - 1 \right) d_t. \quad (6)$$

Substituting Eq. (6) into Eq. (1), the appropriate version of the Orowan equation for magnesium alloys containing $(0001)_\alpha$ precipitate plates is given as

$$\Delta\tau = \frac{Gb}{2\pi\sqrt{1-\nu} \left(\frac{0.953}{\sqrt{f}} - 1 \right) d_t} \ln \frac{d_t}{b}. \quad (7)$$

2.4. $[0001]_\alpha$ precipitate rods

For circular $[0001]_\alpha$ precipitate rods of uniform diameter d_t and length l_t , where $l_t \gg d_t$, the precipitate number density N_a is given as $N_v l_t$. The cross-section of the $[0001]_\alpha$ rods intersected on the slip plane has a circular shape with a diameter of d_t and a mean planar area of $\pi d_t^2/4$. If the $[0001]_\alpha$ rods have a distribution that defines a triangular column within the matrix phase, Fig. 2(b), then the $[0001]_\alpha$ rods intersected by the slip plane of the

matrix phase will have a triangular distribution on that plane, Fig. 2(c). The contribution of the $[0001]_\alpha$ rods to the CRSS increment is phenomenologically similar to that arising from $(0001)_\alpha$ plates, and is given as

$$\Delta\tau = \frac{Gb}{2\pi\sqrt{1-\nu} \left(\frac{0.953}{\sqrt{f}} - 1 \right) d_t} \ln \frac{d_t}{b}. \quad (8)$$

3. Effects of particle shape and orientation on strengthening

Assuming that the volume of each single precipitate particle remains constant when the particle shape is changed from sphere to plate or rod, then

$$\frac{d_t(\text{plate})}{d_t(\text{sphere})} = \left(\frac{2A}{3} \right)^{1/3} \quad \text{and} \quad \frac{d_t(\text{rod})}{d_t(\text{sphere})} = \left(\frac{2}{3A} \right)^{1/3},$$

where A is the aspect ratio of precipitate plates/rods ($A = d_t/t_t$ for plates; $A = l_t/d_t$ for rods). For identical number density of precipitates per unit volume N_v , comparisons of $N_a(\text{plate/rod})/N_a(\text{sphere})$, Table 1, indicate that a change in

Table 1

Effects of precipitate shape and orientation on precipitate number density N_a

Precipitates	$\frac{N_a(\text{plate/rod})}{N_a(\text{sphere})}$
Basal plates	$0.874 A^{-2/3}$
Prismatic plates	$0.874 A^{1/3}$
$[0001]_\alpha$ Rods	$0.874 A^{2/3}$

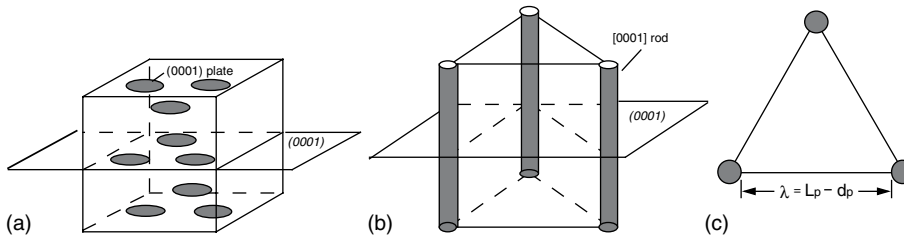


Fig. 2. Arrangement of (a) $(0001)_\alpha$ precipitate plates in a cubic volume of α -Mg matrix phase, and (b) $[0001]_\alpha$ precipitate rods in a triangular prismatic volume of α -Mg. (c) Projection of intersected $(0001)_\alpha$ plates or $[0001]_\alpha$ rods on $(0001)_\alpha$ slip plane of α -Mg. For the purpose of comparison the $(0001)_\alpha$ plates are assumed to have a triangular array distribution in the slip plane of the matrix phase.

precipitate shape from spheres to $[0001]_x$ rods results in the largest increase in N_a .

For a given precipitate volume fraction and number density of precipitates per unit volume of the matrix phase, the variations of the ratio $\Delta\tau(\text{plate}/\text{rod})/\Delta\tau(\text{sphere})$ with the aspect ratio of precipitate plates/rods, for precipitate volume fractions of 0.02 and 0.04, are shown in Fig. 3. To simplify the comparison, the logarithmic terms in the strengthening equations are assumed to be identical. The Orowan increment produced by basal plates is significantly lower than that pro-

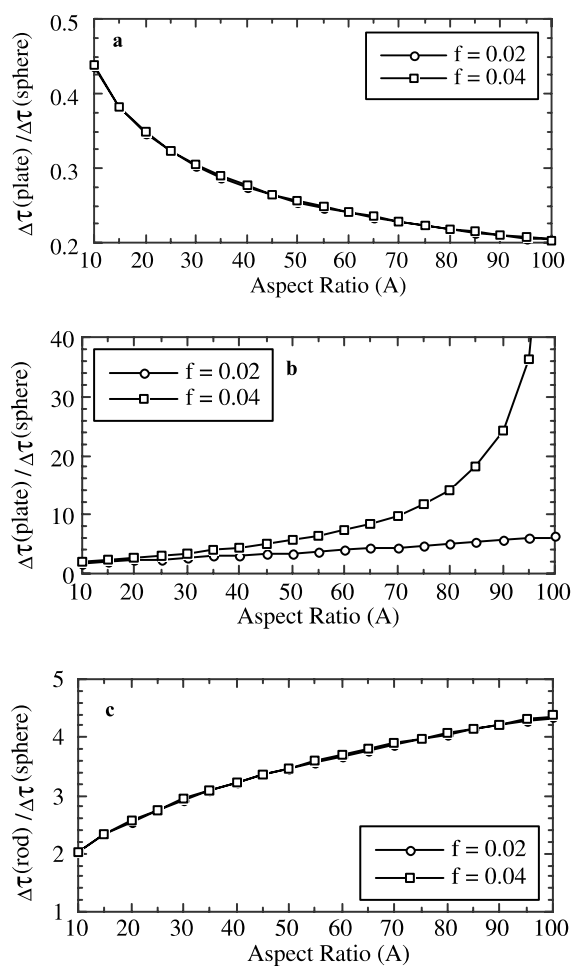


Fig. 3. Variation of ratio $\Delta\tau(\text{plate}/\text{rod})/\Delta\tau(\text{sphere})$ with plate/rod aspect ratio for strengthening attributing to (a) $(0001)_x$ plates, (b) $\{10\bar{1}0\}_x$ or $\{2\bar{1}\bar{1}0\}_x$ prismatic plates, and (c) $[0001]_x$ rods, at precipitate volume fractions of 0.02 and 0.04.

duced by spherical particles, Fig. 3(a), and $\Delta\tau(\text{plate})/\Delta\tau(\text{sphere})$ decreases with an increase in plate aspect ratio. This is because, for a given volume of a precipitate, an increase in plate aspect ratio leads to a reduction in plate thickness and thus a decrease in number density of basal plates per unit area on the slip plane. The ratio $\Delta\tau(\text{plate})/\Delta\tau(\text{sphere})$ is nearly independent of the volume fraction of precipitates, for a precipitate volume fraction in the range 0.02–0.04.

For plate-shaped precipitates formed on prismatic planes of $\alpha\text{-Mg}$, it is interesting to note that the CRSS increment is significantly larger than that produced by spherical particles, and increases substantially with an increase in plate aspect ratio and precipitate volume fraction, Fig. 3(b). For a given arrangement of prismatic plates, the effect of plate aspect ratio on the effective planar inter-plate spacing λ/L_p is demonstrated in Fig. 4. For these prismatic plates, λ/L_p decreases with an increase in the plate aspect ratio, and may reach zero. For $f = 0.04$, the effective planar inter-plate spacing decreases to approximately zero when the aspect ratio of the prismatic plates reaches 105:1. This implies that the precipitate plates will form an essentially closed prismatic volume, defined by the $\{10\bar{1}0\}_x$ or $\{2\bar{1}\bar{1}0\}_x$ plates, when their aspect

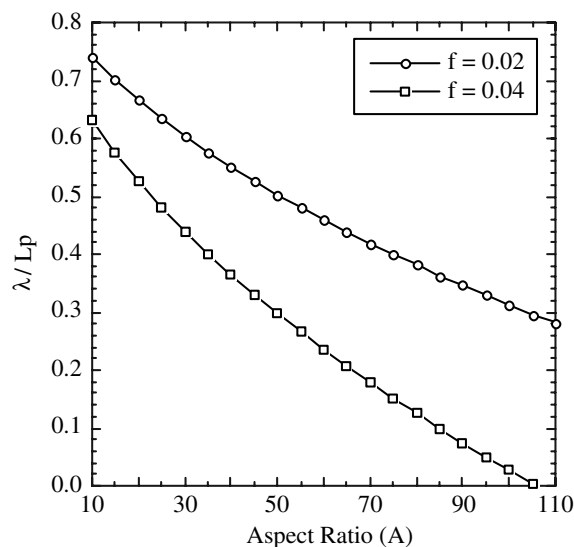


Fig. 4. Variation of ratio λ/L_p with plate aspect ratio for prismatic precipitate plates at volume fractions of 0.02 and 0.04.

ratio exceeds 105:1. Provided that the plates resist shearing by gliding dislocations, the dislocations generated within the prismatic volume cannot escape, and in the limit the theoretical Orowan increment in CRSS approaches infinity. In practice, accumulation of dislocations may lead to local stress concentrations exceeding the yield strength of precipitates and precipitate shearing.

The CRSS increment produced by $[0001]_\alpha$ rods is invariably larger than that produced by spherical particles, Fig. 3(c), and increases with an increase in rod aspect ratio. Similar to that observed for basal plates, the ratio $\Delta\tau(\text{rod})/\Delta\tau(\text{sphere})$ is almost independent of the precipitate volume fraction. While the $[0001]_\alpha$ rods are more effective in strengthening than basal plates, the CRSS increment produced by such rods is substantially smaller than that produced by prismatic plates of large aspect ratios.

The variation of effective inter-particle spacing with precipitate number density is shown in Fig. 5 for a precipitate volume fraction of 0.04. For identical volume fractions and number densities of precipitates, prismatic plates are more effective

than basal plates, $[0001]_\alpha$ rods and spherical particles in achieving a reduction in inter-particle spacing when N_v is in the range typically observed in alloys strengthened by plate-shaped precipitates, i.e. $10 - 10^3 \mu\text{m}^{-3}$ [5].

4. Discussion

Although precipitates in the form of $[0001]_\alpha$ rods are most effective in increasing the precipitate number density per unit area of slip plane of the magnesium matrix phase, the plate-shaped precipitates that form on prismatic planes of the matrix phase are most effective in achieving a reduction in inter-particle spacings and thus in dispersion strengthening. For magnesium alloys containing precipitate particles that are resistant to dislocation shearing, the present model of particle strengthening thus predicts that high strength is associated with microstructures containing a high density of intrinsically strong, plate-shaped precipitates with prismatic habit planes and large aspect ratios, and that precipitate plates formed on the basal plane of the matrix phase provide the least effective barrier to gliding dislocations.

A critical examination of the validity of the present geometric model is currently limited by a lack of reliable experimental data on the size, number density and volume fraction of precipitates, the strengthening mechanisms involved, and a rigorous calculation of dislocation line tension for varying precipitate arrays. Nevertheless, examination of existing precipitation-hardened magnesium alloys indicates that the predictions are in broad qualitative agreement with experimental observations. For example, magnesium alloy AZ91 (Mg–9wt.%Al–1wt.%Zn), strengthened primarily by basal plates of the equilibrium $\text{Mg}_{17}\text{Al}_{12}$ phase [4–6], has a low tensile yield strength (~ 120 MPa) [1]. Arguably, this relatively low strength is attributable to the basal orientation of the precipitate plates and perhaps also to the low number density of such plates. Despite that it is now possible to achieve a substantially higher number density of basal plates in other magnesium alloys [9], the strength achievable in those alloys is still substantially lower than that obtained in alloys

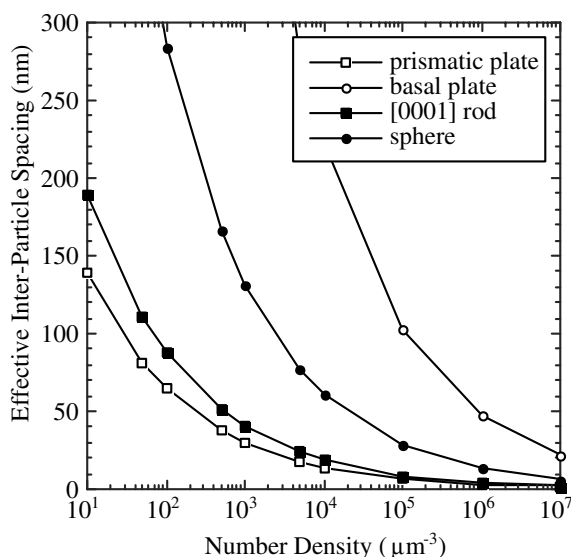


Fig. 5. Variation of effective inter-particle spacing with number density of particles per unit volume of the magnesium matrix at a precipitate volume fraction of 0.04. Plate/rod aspect ratio is 40:1.

strengthened predominantly by prismatic plates, e.g. in WE54 (Mg–5wt.%Y–4wt.%rare earth elements) [1]. The key strengthening precipitate phases in the WE54 alloy are the transition phase β_1 and the equilibrium phase β , both of which form as plates on $\{10\bar{1}0\}_\alpha$ planes of α -Mg [4] and are intrinsically stronger than GP zones and other precipitate phases formed during ageing. It is interesting to note that this alloy exhibits a tensile yield strength of ~ 200 MPa [1] even when the prismatic plates of β_1 and β have small aspect ratios (typically 10:1) and are coarsely distributed in the matrix phase [3]. It is difficult to identify, unambiguously, microstructural factors responsible for the superior strength achieved in this alloy because of the current lack of understanding of precipitate volume fraction size and number density and of strengthening mechanisms. However, it is plausible that the relatively high strength is due primarily to the prismatic orientation of β_1 and β precipitate plates.

Although prismatic precipitate plates of transition or equilibrium phases provide the most effective barriers to dislocations gliding on the basal plane of α -Mg, their aspect ratio and number density are generally much lower than those typical of precipitate plates formed in high strength aluminium alloys. In high strength precipitation-hardened aluminium alloys, the maximum strength is commonly associated with microstructures containing a high density of $\{111\}_\alpha$ precipitate plates of large aspect ratio (typically above 40:1), and these strengthening precipitates are often transition or equilibrium phases [20–22]. Further improvement in the strength of magnesium alloys thus requires an increase in the number density and/or aspect ratio of prismatic plates of transition or equilibrium precipitate phases. Approaches to achieving increases in plate aspect ratio and number density may lie in the use of (i) micro-alloying additions, which partition to either matrix or precipitate phase to improve their coherency, (ii) cold work after solution treatment and prior to ageing, which introduces dislocations to provide heterogeneous nucleation sites for transition and/or equilibrium precipitate phases, and (iii) duplex ageing. Recent studies [23] have demonstrated that cold work after solution treatment and prior to

ageing of alloy WE54 can result in formation of β_1/β plates of larger aspect ratios and a further increase in strength.

5. Conclusions

For magnesium alloys containing identical volume fractions and number densities of shear-resistant precipitates per unit volume, the Orowan increments in yield stress produced by prismatic precipitate plates are invariably larger than those produced by basal precipitate plates, $[0001]_\alpha$ precipitate rods and spherical particles. The yield stress increment produced by prismatic plates increases substantially with an increase in plate aspect ratio. There is a critical value of plate aspect ratio at which the effective inter-particle spacing becomes zero and, if the prismatic plates are assumed to remain shear resistant, then the Orowan increment approaches infinity.

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